Notes: Autor, Dorn, Hanson, and Song (QJE, 2014)

Trade Adjustment: Worker-Level Evidence

Contents

1 Big picture

- **Research question.** What happens to individual workers when their initial industry is hit by rising import competition from China? The paper asks not only whether earnings fall, but also where the adjustment occurs: at the initial employer, across employers, across industries, across regions, and through public transfers.
- **Empirical setting.** The shock is the spectacular rise of Chinese manufacturing exports during 1991–2007. The paper links this shock to U.S. manufacturing industries and follows U.S. workers using longitudinal Social Security Administration earnings records.
- **Unit of analysis.** The analysis is worker-level. Workers are classified by their pre-shock industry of employment, so post-shock industry switching does not mechanically change measured exposure.
- **Identification idea.** The paper uses variation across U.S. manufacturing industries in the growth of import penetration from China. To isolate China-supply-driven shocks, it instruments U.S. import growth from China with Chinese import growth in other high-income countries in the same industries.
- **Main result.** Workers initially in more exposed industries experience lower cumulative earnings and employment over 1992–2007. Comparing the 75th and 25th percentiles of industry trade exposure implies cumulative earnings losses of roughly 40–46 percent of initial annual earnings in the preferred estimates.
- **Adjustment margin.** Losses are concentrated in reduced earnings at the initial employer and in reduced earnings per year of employment. Workers move across firms and sectors, but reallocation does not fully offset lost earnings.
- **Transfer margin.** Exposed workers are more likely to receive Social Security Disability Insurance (SSDI), indicating that trade adjustment partly spills into public transfer programs.
- **Heterogeneity.** Low-wage workers, workers with shorter tenure, and workers with weaker labor-force attachment suffer larger losses. High-wage workers separate more from initial employers but are better able to move out of manufacturing and avoid large earnings losses.
- **Mobility result.** Geographic mobility is limited. The main adjustment happens across employers and industries, not through workers moving across commuting zones.
- **Economic takeaway.** Import competition imposes large, uneven, and long-lived adjustment costs. The costs are not merely local-labor-market averages; they are borne very differently by workers depending on pre-shock skill, tenure, and attachment.

2 Incremental contribution of the paper

2.1 Contribution relative to local-labor-market studies

Earlier China-shock work, especially Autor, Dorn, and Hanson (2013), measured how regional exposure to Chinese imports affected commuting zones. This paper moves from regional averages to individual worker trajectories. That shift matters because local-labor-market regressions cannot say whether the same workers lose earnings, whether workers move to other firms or sectors, whether losses are concentrated among low-wage workers, or whether public transfers absorb part of the shock.

Interpretation: The incremental contribution is a worker-level decomposition of adjustment. Instead of asking whether a region becomes poorer, the paper asks which initially exposed workers lose income, how they respond, and where the losses show up.

2.2 Contribution relative to job-loss studies

The displaced-worker literature documents persistent earnings losses after mass layoffs. This paper differs by identifying the source of displacement pressure: rising import competition from China. Because the shock is industry-level and externally driven, the authors can study a broad set of exposed and less-exposed workers, not only workers who are directly laid off.

Interpretation: The paper bridges trade and labor economics. It shows that trade shocks can create adjustment costs resembling job-loss costs, but the burden is distributed across firms, industries, earnings groups, and transfer systems in ways that standard layoff studies do not reveal.

2.3 Contribution relative to trade theory

Traditional trade models often emphasize eventual reallocation and welfare gains. The paper documents the transitional costs implied by labor-market frictions. The empirical results speak to specific-factors and search-friction models in which workers initially attached to exposed industries do not immediately move to equally productive jobs.

Interpretation: The paper does not estimate the full general-equilibrium welfare effect of trade. Its contribution is measuring the incidence of adjustment costs among workers initially close to the shock.

2.4 Contribution relative to policy debates

Policy debates often ask whether workers displaced by trade are compensated through wage gains elsewhere or through public transfers. This paper quantifies both. It shows that earnings losses are not fully offset by reemployment and that SSDI participation rises, especially for lower-attachment workers.

Interpretation: The results imply that trade adjustment policy should not only target places or industries. Worker characteristics at the start of the shock strongly predict who bears the largest costs.

3 Data and measurement

3.1 China import shock

The central treatment variable is industry-level growth in import penetration from China. For U.S. manufacturing industry j, the baseline measure is

$$\Delta IP_{j,\tau} = \frac{\Delta M^{UC}_{j,\tau}}{Y_{j,1991} + M_{j,1991} - E_{j,1991}}, \tag{1}$$

where $\Delta M^{UC}_{j,\tau}$ is the change in U.S. imports from China in industry j over the period, and the denominator is U.S. domestic absorption in the industry in 1991.

Interpretation: The numerator measures growth in Chinese competitive pressure. The denominator scales the shock by the initial size of the U.S. market, so the variable captures how important the import surge is for the industry's domestic market.

3.2 Instrument for U.S. import growth from China

Because U.S. imports from China may partly reflect U.S. demand shocks, the authors instrument U.S. exposure with Chinese export growth to other high-income countries:

$$\Delta IPO_{j,\tau} = \frac{\Delta M^{OC}_{j,\tau}}{Y_{j,1988} + M_{j,1988} - X_{j,1988}}. \tag{2}$$

Here $\Delta M^{OC}_{j,\tau}$ is the growth of imports from China by other high-income countries in the same industry.

Interpretation: The logic is that China-specific productivity, trade-cost, and export-supply shocks raise Chinese exports to many advanced economies. If import growth is driven by China supply rather than U.S.-specific demand, Chinese exports to other high-income countries should predict U.S. imports from China.

3.3 Trade data and industry concordance

- The paper uses UN Comtrade import data at detailed HS product levels.
- HS products are mapped to SIC manufacturing industries.
- The final industry classification contains 397 manufacturing industries.
- Domestic absorption uses U.S. industry shipments, imports, and exports.
- Other high-income countries used for the instrument include Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland.

Interpretation: The detailed product-to-industry match allows the authors to assign trade shocks at the initial industry level rather than relying only on broad sectors.

3.4 Worker data

- The worker data come from the U.S. Social Security Administration.
- The data follow individual earnings by employer over a long horizon.
- The main outcome window is 1992–2007.

- Workers are assigned to industries based on their initial employment before the shock.
- The baseline sample focuses on workers with high labor-force attachment: workers with at least full-time minimum-wage earnings in each year from 1988 to 1991.
- The extended sample includes workers with weaker attachment, requiring positive earnings in at least one year between 1987–1989 and one year between 1990–1992.

Interpretation: The administrative data make it possible to measure cumulative earnings and firm transitions over many years. The baseline sample avoids confounding trade adjustment with very weak pre-shock labor-market attachment, while the extended sample reveals how more vulnerable workers respond.

3.5 Outcome variables

The main outcome is cumulative post-shock earnings normalized by initial annual earnings:

$$\widetilde{E}_{ij\tau} = \frac{\sum_{t=1992}^{2007} E_{ijt}}{\overline{E}_{i,1988-1991}}. \tag{3}$$

The paper also studies:

- years with positive earnings;
- earnings per year conditional on positive earnings;
- years in which wage earnings, self-employment income, SSDI benefits, or no recorded income are the main income source;
- cumulative earnings by employer, industry, sector, and location.

Interpretation: Normalization by pre-shock earnings makes coefficients interpretable in units of initial annual income. It also avoids the problem that log earnings are undefined for workers with zero earnings in some years.

3.6 Mobility and reallocation variables

The paper decomposes worker outcomes according to whether post-shock earnings occur:

- at the initial employer;
- at another firm in the same two-digit industry;
- at another firm in a different two-digit industry within manufacturing;
- outside manufacturing or in firms with missing industry information;
- in the initial commuting zone or another commuting zone.

Interpretation: This decomposition is central because it distinguishes income losses from job churn. A worker may leave the initial firm and still avoid earnings losses if reallocation is smooth. The paper shows reallocation is often incomplete and unequal across worker types.

3.7 Controls

The preferred specifications include:

- birth-year dummies;

- gender, race, and foreign-born indicators;
- employment history controls, including tenure, experience, and firm size;
- earnings history controls, including average pre-shock log wage and wage trends;
- initial industry import levels;
- manufacturing-sector dummies;
- industry characteristics such as computer investment, high-tech capital share, capital/value added, production-worker share, wage levels, offshoring exposure, and pre-1991 industry trends.

Interpretation: These controls address the concern that China exposure is correlated with pre-existing industry decline, technological change, or worker skill differences.

4 Empirical model and identification

4.1 Economic motivation: labor-market frictions

The paper's motivating model is a two-sector intuition. A foreign productivity shock reduces demand for the U.S. trade-exposed sector. If labor is frictionlessly mobile, similarly skilled workers should reallocate and earnings differences across initial industries should disappear quickly. If workers have firm-specific or industry-specific human capital, search frictions, or geographic frictions, initially exposed workers will suffer cumulative earnings losses during the transition.

Interpretation: The paper's effects are not the full welfare gains or losses from trade. They are the private earnings and employment consequences for workers initially attached to exposed industries.

4.2 Baseline regression

The baseline regression is

$$\widetilde{E}_{ij\tau} = \alpha_0 + \alpha_1 \Delta IP_{j\tau} + \alpha_2 IP_{j,1991} + X'_{ij,0}\alpha_3 + Z'_{j,0}\alpha_4 + \varepsilon_{ij\tau}. \tag{4}$$

Here i indexes workers, j indexes the initial industry, $X_{ij,0}$ is a vector of worker-level controls, and $Z_{j,0}$ is a vector of industry controls.

Interpretation: The coefficient α_1 measures how much cumulative earnings change for workers initially in industries with larger growth in China import penetration.

4.3 Instrumental variables design

The endogenous regressor is U.S. import penetration growth from China. The instrument is Chinese import growth in other high-income countries. The first-stage logic is:

$$\Delta IP^{US}_{j\tau} = \pi_0 + \pi_1 \Delta IP^{Oth}_{j\tau} + controls + u_{j\tau}. \tag{5}$$

The second stage replaces U.S. import penetration growth with its predicted China-supply-driven component.

Interpretation: The identifying assumption is that Chinese export growth to other high-income countries is correlated with China supply shocks but not with unobserved U.S. industry-specific labor-demand shocks, conditional on controls.

4.4 Threats to identification

- **U.S. demand shocks.** Rising U.S. imports from China might reflect strong U.S. demand rather than Chinese supply. The other-country instrument addresses this.
- **Technology shocks.** Labor-intensive industries might decline due to automation rather than China. The paper controls for computer investment, high-tech equipment, capital intensity, and pre-trends.
- **Pre-existing industry decline.** Industries hit by China may have already been declining. The falsification test asks whether future China exposure predicts 1976–1991 outcomes.
- **Worker sorting.** Workers may anticipate industry shocks. The instrument uses 1988 industry affiliation to reduce bias from sorting immediately before 1991.
- **Regional exposure.** A worker may be affected by the local labor market's exposure, not just his industry exposure. Later specifications control for commuting-zone exposure.

Interpretation: The paper's credibility comes from combining an external instrument, rich pre-shock controls, falsification tests, and decompositions by firm, industry, and region.

4.5 What the estimates do and do not identify

- The estimates identify differential outcomes for workers initially employed in more versus less exposed industries.
- They do not identify economy-wide effects on all workers or general-equilibrium changes in wages for broad skill groups.
- If China trade lowers wages for all low-skill workers everywhere, the design would not capture the full effect because it compares workers across initial industries.
- The estimates are best interpreted as the incidence of adjustment costs among initially exposed workers.

Interpretation: This distinction is important: the paper measures who bears the transitional costs of exposure, not the aggregate welfare effect of trade.

5 Detailed results by section

5.1 Section II: Industry trade shocks and worker data

The paper first documents the scale and distribution of the China shock. Chinese manufacturing exports and U.S. import penetration from China rise sharply after the early 1990s, with acceleration around China's WTO entry. Exposure is concentrated in labor-intensive manufacturing sectors such as apparel, textiles, footwear, furniture, and toys.

Interpretation: The shock is large enough to matter and uneven enough across industries to support empirical comparisons. The concentration in production-worker-intensive industries foreshadows the heterogeneity results: low-wage and lower-skill workers are more exposed and have fewer adjustment options.

5.2 Section III.A: Cumulative earnings, years worked, and earnings per year

The baseline estimates show a large negative effect of China exposure on cumulative earnings. In Table II, the preferred 2SLS coefficient is about -6.864. With the 75th–25th percentile exposure gap among manufacturing workers of about 6.68 percentage points, the implied cumulative earnings loss is about 46 percent of initial annual earnings.

Table III decomposes this loss. The coefficient on years with positive earnings is negative but not statistically precise, while the coefficient on earnings per year is negative and significant. Thus, losses come more from reduced earnings in years of work than from many additional full years with zero earnings.

Interpretation: Trade exposure depresses both employment and wage/earnings intensity, but the main baseline channel is lower earnings conditional on working, not simply complete nonemployment.

5.3 Section III.B: Falsification using pre-1991 outcomes

The paper uses 1976–1991 outcomes to test whether industries later hit by China were already on a path of lower earnings. Future China exposure does not significantly predict past cumulative earnings, years worked, or earnings per year.

Interpretation: This falsification test supports the view that the main results reflect post-1991 import competition rather than a long-run secular trend that started before the China shock.

5.4 Section III.C: Reallocation across firms, industries, sectors, and regions

Table IV decomposes earnings and employment by employer and industry. Workers in more exposed industries lose earnings at their initial firm. They gain some employment and earnings elsewhere, but the offsets are incomplete. Figure III shows the dynamics: job churn rises over time, with increasing losses at the initial employer and increasing employment outside the original non-employment state.

Interpretation: The labor market adjusts through job churn, but not costlessly. Workers move, yet their new jobs do not fully replace lost earnings. This is evidence of firm-specific or industry-specific rents/human capital and search frictions.

5.5 Section III.D: Geographic mobility

Table V shows that geographic mobility is not a major adjustment channel. More exposed workers lose earnings in the initial commuting zone and also in other locations, with little evidence that moving across regions offsets losses.

Interpretation: Trade adjustment is not mainly a migration story. Workers may change employers and sectors, but large-scale movement across local labor markets does not appear to undo the income effects.

5.6 Section III.E: Transfer programs and disability insurance

Tables VI and VII show that import competition increases reliance on SSDI. In the main sample, exposure raises years in which SSDI is the primary income source. In the extended sample with lower labor-force attachment, the SSDI response is larger.

Interpretation: The cost of trade adjustment partly enters the public budget. Workers who cannot maintain stable earnings may transition into disability benefits, especially if they have weaker initial attachment to work.

5.7 Section IV: Heterogeneity across workers

Tables VIII and IX show that trade adjustment is highly unequal.

- Workers with shorter tenure experience larger losses than workers with longer tenure.
- Older workers face more difficult adjustment in some specifications.
- Low-wage workers experience large cumulative earnings losses and lower earnings per year.
- High-wage workers lose employment at the initial firm but are better able to move outside manufacturing and avoid large earnings losses.

Interpretation: Initial labor-market position determines exposure to adjustment costs. High-wage workers appear more mobile across sectors; low-wage workers churn within manufacturing, where they remain exposed to subsequent trade shocks.

5.8 Section V: Alternative exposure measures and regional controls

Table X adds local-labor-market exposure alongside industry exposure. Industry exposure remains important, while commuting-zone exposure also matters but is smaller because CZ exposure varies less than industry exposure. Table XI shows robustness to alternative import-exposure measures, including gravity residuals, low-income country imports, worker-equivalent units, and imports from Mexico/CAFTA.

Interpretation: The preferred result is not an artifact of one exposure measure. The robustness tests support the interpretation that China import competition has persistent worker-level consequences.

6 Tables and figures

6.1 Figure I and Figure II: the scale and distribution of the China shock

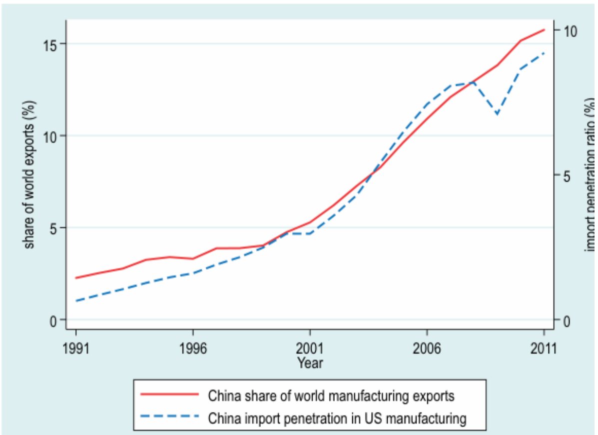

Figure I: China export share and U.S. manufacturing import penetration

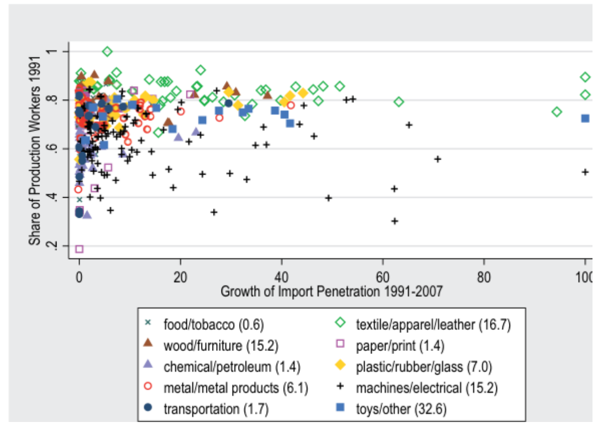

Figure II: industry exposure versus production-worker intensity

Read:

- Figure I shows China's share of world manufacturing exports and the China import-penetration ratio in U.S. manufacturing rising steeply after 1991.
- The increase accelerates in the 2000s, consistent with China's accession to the WTO and rapid manufacturing productivity growth.
- Figure II shows that exposure is concentrated in labor-intensive industries with high production-worker shares.
- Sectors such as toys, apparel, leather, textiles, furniture, and wood products show especially large increases in import penetration.

Economic message: The shock is both large and uneven. This creates empirical variation and implies that production-worker-intensive industries are likely to bear the greatest adjustment costs.

6.2 Table I: descriptive statistics

Read:

- The main sample contains 508,129 workers; 105,625 are manufacturing workers.
- Among manufacturing workers, mean trade exposure is 7.72 percentage points, with a 75th–25th percentile gap of 7.30 minus 0.62.
- The average high-attachment manufacturing worker has positive earnings in about 14.3 of 16 years.
- Manufacturing workers spend about 0.43 years with SSDI as the main income source over the outcome window.

Economic message: The sample is large, longitudinal, and contains enough exposure dispersion to estimate meaningful worker-level trade adjustment costs.

TABLE I

DESCRIPTIVE STATISTICS

	Main Sample		Extended Sample	
	All Workers	Manuf Workers	All Workers	Manuf Workers
Panel A: trade exposure, 1991–2007				
(Δ imports from China to U.S.)/U.S. consumption$_{91}$	1.60	7.72	1.30	6.30
	(7.05)	(13.85)	(5.85)	(11.59)
P90, P10 interval	[2.72, 0.00]	[24.74, 0.06]	[1.99,0.00]	[18.75, 0.04]
P75, P25 interval	[0.00, 0.00]	[7.30, 0.62]	[0.00,0.00]	[6.28, 0.38]
(1991 imports from China to U.S.)/U.S. consumption$_{91}$	0.11	0.54	0.11	0.48
	(0.94)	(2.00)	(0.88)	(1.78)
Panel B: main outcome variables, 1992–2007				
100*cumulative earnings (in mult of avg annual wage 88–91)	1,918.4	1,808.9	n/a	n/a
	(1,181.7)	(1,025.7)		
100*number of years with earnings > 0	1422.3	1428.6	1326.0	1355.8
	(342.1)	(335.7)	(428.3)	(403.6)
100*cumul earn/yrs with earn > 0 (in mult of avg ann wage 88–91)	130.0	122.4	n/a	n/a
	(69.9)	(59.7)		
100*number of years with main income from SSDI	34.3	43.3	44.4	50.9
	(168.3)	(188.6)	(200.0)	(211.4)
100*number of years with main income from self-employment	46.4	37.7	59.8	47.0
	(175.9)	(152.2)	(200.4)	(170.2)
Panel C: worker characteristics in 1991				
Female	0.431	0.312	0.475	0.359
Nonwhite	0.207	0.200	0.236	0.235
Foreign-born	0.077	0.086	0.085	0.097
Employed in manufacturing	0.208	1.000	0.173	0.836
Tenure 0–1 years	0.269	0.236	0.418	0.425
Tenure 2–5 years	0.368	0.355	0.301	0.283
Tenure 6–10 years	0.169	0.187	0.129	0.134
Tenure 11+ years	0.194	0.221	0.153	0.158
Firm Size 1–99 employees	0.231	0.153	0.257	0.199
Firm Size 100–999 employees	0.237	0.289	0.231	0.285
Firm Size 1,000–9,999 employees	0.246	0.290	0.210	0.251
Firm Size 10,000+ employees	0.286	0.267	0.302	0.265
Average log wage 1988–1991	10.46	10.55	9.23	9.60
Sample size	508,129	105,625	880,465	181,900

Table I: descriptive statistics

TABLE II

IMPORTS FROM CHINA AND CUMULATIVE EARNINGS, 1992–2007: OLS AND 2SLS ESTIMATES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	OLS				2SLS				
(Δ China imports)/U.S. consumption$_{91}$	−2.939*	−5.726**	−6.017**	−6.099**	−6.883**	−6.436**	−5.695*	−6.728**	−6.864**
	(1.150)	(1.425)	(1.475)	(1.395)	(1.351)	(2.450)	(2.326)	(2.392)	(2.477)
Birth year dummies	yes	yes	yes	yes	yes	yes	yes	yes	yes
Demographic controls			yes	yes	yes	yes	yes	yes	yes
Employment history				yes	yes	yes	yes	yes	yes
Earnings history					yes	yes	yes	yes	yes
Initial import levels						yes	yes	yes	yes
10 mfg industry dummies							yes	yes	yes
Industry characteristics								yes	yes
Pretrends ind emp and wage									yes

Notes. Dependent variable: 100 × earnings 1992–2007 (in multiples of initial annual wage). $N = 508{,}129$. All regressions include a constant and a full set of birth year dummies. Demographic controls in column (3) include dummies for female, nonwhite, and foreign-born. Employment history controls in column (4) include dummies for tenure at 1991 firm (0–1, 2–5, 6–10 years), experience (4–5, 6–8, 9–11 years), and size of 1991 firm (1–99, 100–999, 1,000–9,999 employees). Earnings history controls in column (5) include the worker's annual log wage averaged over 1988–1991, an interaction of intial wage with age, and the change in log wage between 1988 and 1991, as well as the level and trend of the 1991 firm's log mean wage for the period 1988–1991. Columns (6)–(9) add controls for a worker's 1991 manufacturing industry, starting with initial trade penetration by Chinese and non-Chinese imports in column (6). Column (7) adds dummies for 10 manufacturing subindustries, column (8) adds 1991 levels for employment share of production workers, log average wage, capital/value added, and 1990 levels for computer investment, share of investment allocated to high-tech equipment, and fraction of intermediate goods among imports in 1990; column (9) additionally controls for changes in industry employment share and log average wage level during the preceding 16 years (1976–1991). Robust standard errors in parentheses are clustered on start-of-period three-digit industry. $\sim p \leq .10$, $^{*}p \leq .05$, $^{**}p \leq .01$.

Table II: cumulative earnings, OLS and 2SLS

6.3 Table II and Table III: baseline earnings losses and falsification

TABLE III

IMPORTS FROM CHINA 1992–2007 AND CUMULATIVE EARNINGS, YEARS WITH EARNINGS, AND EARNINGS PER YEAR 1992–2007 AND 1976–1991: 2SLS ESTIMATES

	(1) Cumulative Earnings	(2) Years w/ Earnings > 0	(3) Earnings/ Year
Panel A: treatment period 1992–2007			
(Δ China imports)/U.S. consumption$_{91}$	−6.864**	−0.535	−0.393**
	(2.477)	(0.505)	(0.140)
Panel B: preperiod 1976–1991			
(Δ China imports)/U.S. consumption$_{91}$	−0.432	0.695	−0.064
	(1.996)	(0.856)	(0.112)

Notes. Dependent variables: 100 × cumulative earnings (in multiples of initial annual wage); 100 × years with earnings; 100 × earnings per year of employment (in multiples of initial annual wage), $N = 508{,}792$ in Panel A and $N = 301{,}490$ in Panel B, except $N = 506{,}339$ and $N = 300{,}239$ in column (3), where the dependent variable is not defined for individuals who are never employed during the entire outcome period. Regressions in Panel A include the full control vector from column (9) of Table II. Regressions in Panel B include the same controls except tenure, experience, and firm size; industry-level controls are measured either for 1975 or for 1972 (intermediate imports, computer investment, and high-tech equipment). Robust standard errors in parentheses are clustered on start-of-period three-digit industry. $\sim p \leq .10$, $^{*}p \leq .05$, $^{**}p \leq .01$.

Table III: earnings, years worked, and earnings per year

the sample period: total labor earnings, the number of
with positive labor earnings, earnings per year for year
nonzero earnings, total self-employment income, and tota
benefits received.[20] Table I describes variation in these ou
across workers. For the main sample of workers with hig
force attachment, the average worker had positive labor ea
in 14.2 of the 16 years, cumulatively earned 19.2 times his
average annual wage (measured as the average of the

Baseline regression excerpt

Read:

- Table II column (1) OLS gives a negative coefficient of about −2.939.
- The 2SLS coefficients are larger in magnitude, suggesting OLS is biased toward zero by U.S. demand or other confounds.
- The preferred column (9) coefficient is about −6.864, implying a large cumulative earnings loss for more exposed workers.
- Table III shows that the effect is concentrated in lower earnings per year rather than many additional years with zero earnings.

- Panel B of Table III shows that future China exposure does not predict past 1976–1991 outcomes, supporting the identification strategy.

Economic message: The headline result is large and robust: exposure to Chinese import competition lowers cumulative earnings for workers initially attached to exposed industries.

6.4 Figure III and Table IV: job churning and reallocation

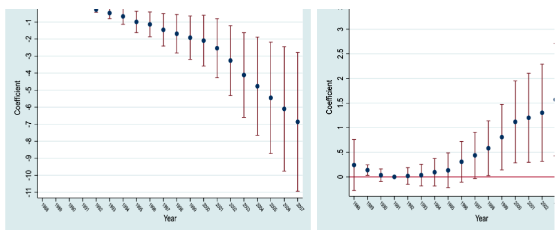

FIGURE III

Cumulative Earnings and Cumulative Job Churning since 1991

Each panel plots regression coefficients and 90% confidence intervals obtained from 20 regressions that relate outcome variables to the 1991–2007 trade exposure of a worker's 1991 industry. The outcome in the left panel is earnings that a worker obtained from 1991 through the year indicated on the *x*-axis, expressed in percentage points average annual earnings in 1988–1991. Coefficients for years prior to 1992 refer to cumulative earnings between the ye the *x*-axis and 1991. The outcome variable in the right panel is the cumulative number of times a worker has change moved between employment and nonemployment between 1991 and the year indicated on the *x*-axis, multiplied by 100 variable takes negative values prior to 1991, such that positive coefficients pre-1991 indicate lower churning for worke industry was trade-exposed. All regressions include the vector of control variables from column (9) of Table I.

Figure III: cumulative earnings and job churning since 1991

TABLE IV

IMPORTS FROM CHINA AND EARNINGS AND EMPLOYMENT BY FIRM, INDUSTRY, AND SECTOR, 1992–2007: 2SLS ESTIMATES

	(1)	(2)	(3)	(4)	(5)	(6)
	All Employers	Same Sector			Other Sector	N/A
Same 2-digit Industry		Yes	Yes	No	No	N/A
Same Firm		Yes	No	No	No	No
Panel A: cumulative earnings (in initial annual wage*100)						
(Δ China imports)/	−6.864**	−9.107**	−2.998~	5.914*	−2.253	1.579*
U.S. consumption$_{91}$	(2.477)	(3.395)	(1.671)	(2.326)	(3.049)	(0.800)
Panel B: cumulative employment (in years*100)						
(Δ China imports)/	−0.535	−6.204*	−2.036	4.654**	1.451	1.600*
U.S. consumption$_{91}$	(0.505)	(2.566)	(1.278)	(1.655)	(1.983)	(0.625)
Panel C: earnings per year of emp (in initial annual wage*100)						
(Δ China imports)/	−0.393**	−0.283	−0.543~	−0.551~	−0.655*	−0.606*
U.S. consumption$_{91}$	(0.140)	(0.093)	(0.304)	(0.300)	(0.291)	(0.245)

Notes. Dependent variables: 100 × cumulative earnings; 100 × years with earnings; 100 × earnings per year of employment. N = 508,129 in Panels A and B. N = 506,339, 424,027, 155,993, 263,158, 112,002, 119,989 in columns (1)–(6) of Panel C. Column (6) measures employment and earnings in firms with missing industry information. A large majority of these firms are new firms that have been incorporated

Table IV: earnings and employment by firm, industry, and sector

Read:

- Figure III shows that cumulative earnings effects grow over time rather than appearing only immediately after the shock.
- More exposed workers lose earnings and employment at their initial firm.
- Table IV shows partial offsets through employment at other firms and sectors, but the earnings offsets are incomplete.
- Workers spend less time at the initial employer and more time elsewhere in manufacturing or outside manufacturing.

Economic message: Trade adjustment involves real job churn. However, churn is not a frictionless reallocation process; new jobs do not restore the lost earnings of many displaced or exposed workers.

6.5 Figure IV and Table V: persistence and geographic mobility

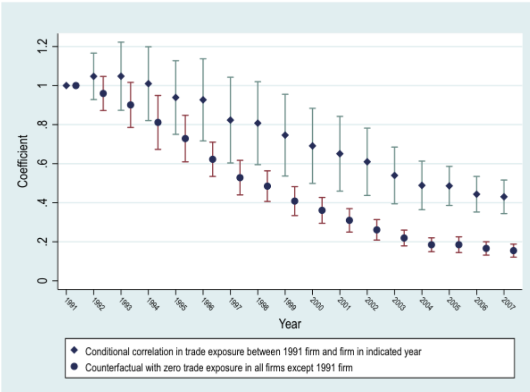

FIGURE IV

Persistence of Trade Exposure since 1991

Figure IV: persistence of trade exposure since 1991

TABLE V

IMPORTS FROM CHINA AND EARNINGS AND EMPLOYMENT BY GEOGRAPHIC LOCATION, 1994–2007: 2SLS ESTIMATES

	(1) All CZ	(2) Initial CZ	(3) Other CZ	(4) N/A CZ
Panel A: cumulative earnings (in initial annual wage*100)				
Δ China imports)/U.S. consumption$_{91}$	−6.649**	−3.558*	−2.307~	−0.785
	(2.417)	(1.396)	(1.281)	(0.542)
Panel B: cumulative employment (in years*100)				
Δ China imports)/U.S. consumption$_{91}$	−0.653	−0.008	−0.491	−0.154
	(0.513)	(0.599)	(0.539)	(0.253)
Panel C: earnings/year (in initial annual wage*100)				
Δ China imports)/U.S. consumption$_{91}$	−0.418**	−0.344**	−0.741*	−1.010*
	(0.152)	(0.124)	(0.369)	(0.398)

Notes. Dependent variables: 100 × cumulative earnings; 100 × years with earnings; 100 × earnings per year of employment. N = 508,129, except smaller samples in Panel C. Column (1) shows aggregate results for earnings and employment. Columns (2)–(4) subdivide results into employment and earnings obtained while residing in the 1993 CZ of residence (column (2)), in all CZs other than the 1993 CZ of residence (column (3)), and in CZs than cannot be classified because either the 1993 location or subsequent location of the worker is unknown (column (4)). All regressions include the full control vector from column (9) of Table II, and a dummy for the 6% of workers whose 1993 CZ is unknown. Robust standard errors in

Table V: earnings and employment by commuting zone

Read:

- Figure IV shows that workers remain correlated with their initial exposure for many years, even after changing jobs.
- Table V indicates that geographic mobility does not offset earnings losses.
- Exposed workers lose earnings in the initial commuting zone and also do not gain enough in other commuting zones.

Economic message: The persistence of exposure helps explain why adjustment costs last. Workers do not instantly escape their initial industry's shock through mobility.

6.6 Tables VI and VII: nonwage income and SSDI

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TABLE VI

IMPORTS FROM CHINA AND NONWAGE INCOME, 1992–2007: 2SLS ESTIMATES

	(1) Wage Earnings	(2) Self-Emp. Income	(3) SSDI Income	(4) No Record Income
Panel A: Cumulative Years with Main Income from Indicated Source				
(Δ China imports)/U.S. consumption$_{91}$	−0.493	−0.127	0.562*	0.057
	(0.498)	(0.207)	(0.268)	(0.367)
Panel B: Cumulative Income from Indicated Source (in Initial Annual Wage*100)				
(Δ China imports)/U.S. consumption$_{91}$	−6.864**	−0.007	0.352*	n/a
	(2.477)	(0.348)	(0.164)	
Panel C: Years with SSDI Income: Intensive vs Extensive Margin				
	100*Dummy (Yrs SSDI > 0)		100*Yrs SSDI if Yrs SSDI > 0	

Table VI: nonwage income and SSDI

TABLE VII

IMPORTS FROM CHINA AND CUMULATIVE YEARS OF EMPLOYMENT AND SSDI RECEIPT BY WORKER SAMPLE: 2SLS ESTIMATES

	(1)	(2)	(3)	(4)	(5)
	Cumulative Years with Main Income from				
	Wage Earnings	Self-Employmt	SSDI Benefits	No Income	100*Dummy (Yrs SSDI > 0)
Panel A: main sample of high LF attachment workers					
(Δ China imports)/U.S. consumption$_{91}$	−0.613	−0.152	0.438*	0.327	0.056*
	(0.415)	(0.157)	(0.202)	(0.330)	(0.026)
			N = 508,129		
Panel B: extended sample with high and low LF attachment workers					
(Δ China imports)/U.S. consumption$_{91}$	−0.991~	−0.694*	1.137**	0.548	0.125**
	(0.569)	(0.301)	(0.431)	(0.426)	(0.043)
			N = 880,465		

Notes. Dependent variables: 100 × cumulative years with income from indicated sources and 100 × dummy for SSDI income, 1993–2007. Panel A includes workers who earned at least the equivalent of a full-time minimum wage job in each of the four years 1988 to 1991. Panel B adds workers with low annual incomes or interrupted careers who were employed at least in one year during 1987–1989, and one year during 1990–1992. All regressions include the full vector of control variables from column (9) of Table II, except that all industry- and firm-related variables are averaged over those years in which a worker was employed during 1990 to 1992. Robust standard errors in parentheses are clustered on start-of-period three-digit industry. ~$p \leq .10$, *$p \leq .05$, **$p \leq .01$.

Table VII: main and extended worker samples

Read:

- Table VI shows a positive effect of trade exposure on years with SSDI as the main income source.

- The intensive/extensive decomposition suggests an increased probability of receiving SSDI rather than only more years conditional on receipt.
- Table VII shows a stronger SSDI response in the extended sample, which includes lower-attachment workers.
- The extended sample also shows larger losses in labor earnings and self-employment income.

Economic message: Trade shocks can shift costs from private earnings to public transfer systems. The effect is strongest for workers who start with weaker attachment to work.

6.7 Tables VIII and IX: heterogeneity by tenure, age, and wage

TABLE VIII

IMPORTS FROM CHINA AND EARNINGS AND EMPLOYMENT BY FIRM TENURE AND AGE, 1992–2007: 2SLS ESTIMATES

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Outcomes: Overall and by Employer				Outcomes: Overall and by Employer			
	All Firms	Initial Firm	Same Sector	Oth. Sect./NA	All Firms	Initial Firm	Same Sector	Oth. Sect./NA
	Panel A1: workers w/ firm tenure < 5 years				Panel B1: workers born in 1957–1970			
Cumul. earnings	−15.06**	−15.53**	−1.28	1.75	−8.71*	−8.98**	2.45	−2.18
	(5.53)	(5.77)	(5.02)	(6.76)	(3.52)	(3.40)	(3.37)	(4.09)
Cumul. yrs. emp.	0.84	−9.03*	1.99	7.88~	0.94	−4.43*	2.49	2.88
	(0.99)	(3.88)	(3.14)	(4.63)	(0.64)	(2.09)	(1.78)	(2.47)
Cumul. earn./year	−0.95**	−0.60**	−1.65**	−1.22*	−0.61**	−0.56**	−1.06*	−0.68*
	(0.34)	(0.20)	(0.60)	(0.51)	(0.23)	(0.18)	(0.50)	(0.31)
	Panel A2: workers w/ firm tenure ≥ 5 years				Panel B2: workers born in 1943–1956			
Cumul. earnings	−1.74~	−4.99*	0.06	3.20	−5.63**	−8.39**	−0.08	2.84
	(0.94)	(2.08)	(0.99)	(1.98)	(2.03)	(3.28)	(1.94)	(2.67)
Cumul. yrs. emp.	−1.27*	−4.18*	0.17	2.74	−1.38~	−6.84*	1.42	4.04
	(0.59)	(1.86)	(0.76)	(1.68)	(0.76)	(2.80)	(1.43)	(2.49)
Cumul. earn./year	−0.05	−0.07	0.01	0.13	−0.25**	−0.09	−0.47*	−0.25
	(0.06)	(0.06)	(0.08)	(0.11)	(0.09)	(0.08)	(0.21)	(0.17)

Notes. Dependent variables: 100 × cumulative earnings (in multiple of initial annual wage); 100 × years with earnings; 100 × earnings per year of employment (in multiples of initial annual wage). N = 293,816, 214,313, 236,136, 249,921 for Panels A1, A2, B1 and B2, except smaller samples for cumulative earnings per year of employment. Columns (4) and (8) report outcomes at firms outside the initial sector and at firms with missing industry information (most of which have been incorporated between 2000 and 2007). All regressions include a constant and the full vector of control variables from column (9) of Table II. Robust standard errors in parentheses are clustered on start-of-period three-digit industry. ~$p \leq .10$, *$p \leq .05$, **$p \leq .01$.

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Table VIII: heterogeneity by firm tenure and age

TABLE IX

IMPORTS FROM CHINA AND EARNINGS AND EMPLOYMENT BY SUBGROUP OF WORKERS: 2SLS ESTIMATES

	(1)	(2)	(3)	(4)
	Worker Outcomes: Overall and by Employer			
	All Firms	Initial Firm	Oth Firm, Same Sector	Other Sector/NA
Panel A: initial wage in bottom tercile of cohort				
Cumulative earnings	−18.27**	−8.78*	0.85	−10.34*
	(5.76)	(3.84)	(5.21)	(4.52)
Cumulative years employed	−0.20	−4.10~	4.50~	−0.60
	(1.30)	(2.23)	(2.73)	(2.68)
Cumulative earnings/year	−1.16**	−0.44~	−2.71**	−1.34**
	(0.34)	(0.24)	(0.85)	(0.45)
Panel B: initial wage in middle tercile of cohort				
Cumulative earnings	−9.93**	−8.50**	2.73	−4.16
	(3.65)	(3.27)	(2.38)	(3.17)
Cumulative years employed	−0.37	−4.43*	2.84~	1.22
	(0.65)	(2.03)	(1.65)	(1.88)
Cumulative earnings/year	−0.61**	−0.51**	−0.67~	−0.67*
	(0.22)	(0.17)	(0.35)	(0.31)
Panel C: initial wage in top tercile of cohort				
Cumulative earnings	−0.22	−9.15~	2.24	6.69
	(2.07)	(4.69)	(2.36)	(5.38)
Cumulative years employed	−0.41	−7.79~	1.10	6.28
	(0.55)	(4.29)	(1.59)	(4.29)
Cumulative earnings/year	0.03	−0.04	0.07	0.03
	(0.14)	(0.12)	(0.29)	(0.22)

Notes. Dependent variables: 100 × cumulative earnings (in multiples of initial annual wage); 100 × years with earnings; 100 × earnings per year of employment (in multiples of initial annual wage). N = 169,386.

Table IX: heterogeneity by initial wage tercile

Read:

- Table VIII shows that workers with shorter tenure suffer larger earnings losses than workers with long tenure.
- Low-tenure workers are less protected by firm-specific human capital, seniority, or internal labor markets.
- Table IX shows that low-wage workers suffer large cumulative earnings losses and lower earnings per year.
- High-wage workers lose time at the initial firm but offset losses more effectively by moving outside manufacturing.

Economic message: Adjustment costs are not uniform. Low-wage and low-tenure workers bear more persistent losses, while high-wage workers appear to have better outside options.

TABLE X

EFFECT OF EXPOSURE TO CHINESE IMPORTS AT THE INDUSTRY AND COMMUTING ZONE LEVEL ON EARNINGS AND EMPLOYMENT: 2SLS ESTIMATES

	(1)	(2)	(3)	(4)	(5)	(6)
		Outcomes Measured at				
		I. All CZ		II. Initial CZ	III. Other CZ	IV. N/A CZ
Panel A: cumulative earnings						
(Δ Imports from China to U.S.)/U.S. consumption$_{91}$	−6.65**		−6.07**	−3.31**	−2.09~	−0.66
	(2.42)		(2.16)	(1.24)	(1.25)	(0.59)
CZ avg of [(Δ imports from China to U.S.)/U.S. consumption$_{91}$]		−20.30**	−19.68**	−16.88**	−0.56	−2.24
		(6.72)	(6.79)	(4.59)	(4.88)	(1.38)
Panel B: cumulative years of employment						
(Δ Imports from China to U.S.)/U.S. consumption$_{91}$	−0.65		−0.65	−0.14	−0.38	−0.14
	(0.51)		(0.50)	(0.65)	(0.58)	(0.31)
CZ avg of [(Δ imports from China to U.S.)/U.S. consumption$_{91}$]		2.26~	2.33~	0.42	1.15	0.76
		(1.23)	(1.22)	(3.13)	(2.74)	(0.71)
Panel C: earnings per year of employment						
(Δ Imports from China to U.S.)/U.S. consumption$_{91}$	−0.42**		−0.37**	−0.30**	−0.71*	−0.97**
	(0.15)		(0.14)	(0.12)	(0.36)	(0.36)
CZ avg of [(Δ imports from China to U.S.)/U.S. consumption$_{91}$]		−1.66**	−1.62**	−1.59**	−1.02~	−1.33~
		(0.58)	(0.59)	(0.52)	(0.58)	(0.81)

Notes. Dependent variable: 100 × cumulative earnings, years of employment, or earnings/year, 1994–2007. $N = 508{,}129$, except smaller samples in Panel C. The CZ average of import exposure for worker i in CZ j is defined as the average trade exposure of all other workers $-i$ located in CZ j, according to the industry employment structure obtained from the 1990 County Business Patterns. Workers' 1991 CZ of residence is not observed in the data and thus imputed based on available location and employment information. For most workers, it corresponds to the observed 1993 CZ of residence. All regressions include the full control vector from column (9) of Table II and regressions in columns (2)–(6) include a dummy for observations with missing 1993 CZ and dummies for 1991 census divisions. Robust standard errors in parentheses are two-way clustered on start-of-period three-digit industry and on imputed 1991 CZ of residence. ~$p \le .10$, *$p \le .05$, **$p \le .01$.

Table X: industry and commuting-zone exposure

TABLE XI

ALTERNATIVE MEASURES OF IMPORT EXPOSURE: 2SLS ESTIMATES

Alternative Measure of Trade Exposure	(1) Cumulative Earnings	(2) Years w/ Earn > 0	(3) Earn/ Year	Alternative Measure of Trade Exposure	(4) Cumulative Earnings	(5) Years w/ Earn > 0	(6) Earn/ Year
	Panel A: 2SLS main results				Panel E: 2SLS (instr: Chn-OTH trade)		
Δ Import penetration, using imports from China	−6.864**	−0.535	−0.393**	Δ Import penetration, using China imports to U.S. and other markets	−5.616**	−0.493	−0.321**
	(2.477)	(0.505)	(0.140)		(2.029)	(0.442)	(0.113)
	Panel B: Reduced form OLS				Panel F: 2SLS (instr: Chn-OTH trade)		
Δ China–U.S. productivity differential (gravity residual)	−4.684**	−0.383	−0.270**	Δ Net import penetration, using China imports	−4.170**	−0.761*	−0.204*
	(1.494)	(0.378)	(0.090)		(1.410)	(0.360)	(0.081)
	Panel C: 2SLS (instr: Low-OTH trade)				Panel G: 2SLS (instr: Chn-OTH trade)		
Δ Import penetration, using all low-income country imports	−6.836**	−0.489	−0.396**	Δ Net imports from China in worker-equivalent units	−5.658**	−0.790	−0.301**
	(2.409)	(0.486)	(0.138)		(1.965)	(0.497)	(0.112)
	Panel D: 2SLS (instr: Mex-OTH trade)				Panel H: 2SLS (instr: Chn-OTH trade)		
Δ Import penetration, using imports from Mexico/CAFTA	−0.118	1.583	0.029	Δ Import penetration, using China imp. adjusted for imported inputs	−6.499**	−0.568	−0.368**
	(6.688)	(2.160)	(0.411)		(2.386)	(0.514)	(0.134)

Notes. Dependent variables: 100 × cumulative earnings; 100 × years with earnings (in multiples of initial annual wage); 100 × earnings per year of employment (in multiples of initial annual wage), 1992–2007. $N = 508{,}129$ in columns (1)–(2) and (4)–(5), $N = 506{,}339$ in columns (3) and (6). Panel A repeats results from Tables II and III. The mean (and standard deviation) of trade exposure among manufacturing workers is 1.02 (3.11) in Panel B, 8.56 (14.94) in Panel C, 3.20 (5.54) in Panel D, 8.62 (15.22) in Panel E, 6.14 (13.96) in Panel F, 5.91 (13.87) in Panel G, and 5.77 (12.69) in Panel H. All models include the full vector of control variables from column (9) of Table II, except that start-of-period import exposure levels are adjusted to reflect the alternative definitions of import exposure measures where feasible. Robust standard errors in parentheses are clustered on start-of-period three-digit industry. ~$p \le .10$, *$p \le .05$, **$p \le .01$.

Table XI: alternative measures of import exposure

6.8 Tables X and XI: region controls and alternative exposure measures

Read:

- Table X shows that industry-level exposure remains important even when local-labor-market exposure is included.
- The comparison implies that workers are affected by both the industry shock and the regional economic environment, but industry exposure is central in this worker-level design.
- Table XI shows that the main results survive alternative exposure definitions, including gravity-based exposure and other country comparisons.
- Results are much weaker when using imports from Mexico/CAFTA, supporting the interpretation that the China shock is distinctive.

Economic message: The results are robust to alternative ways of measuring exposure and do not simply reproduce a generic import or regional decline pattern.

7 Short summary

- The paper studies worker-level adjustment to rising Chinese import competition in the United States.
- Workers are assigned exposure based on their initial pre-shock industry, which avoids post-shock sorting bias.
- The main exposure variable is growth in Chinese import penetration from 1991 to 2007.
- The IV strategy instruments U.S. imports from China using Chinese import growth in other high-income countries.
- The baseline sample contains high labor-force-attachment workers observed in SSA administrative data.
- More exposed workers experience lower cumulative earnings over 1992–2007.
- A 75th-versus-25th percentile exposure comparison implies losses around 40–46 percent of initial annual earnings in preferred specifications.
- The earnings loss is driven mainly by lower earnings per year of employment, not just many extra full years out of work.
- Exposed workers leave initial employers and industries, but reallocation only partially offsets earnings losses.
- Geographic mobility is limited and does not undo the losses.

- Trade exposure raises SSDI receipt, especially among weaker-attachment workers.
- Low-wage, low-tenure, and lower-attachment workers suffer the largest losses.
- High-wage workers are more able to move outside manufacturing and avoid large earnings losses.
- Falsification tests show that future China exposure does not predict past outcomes.
- Alternative import-exposure measures support the main interpretation.

8 Conclusion

8.1 Core conclusion

Autor, Dorn, Hanson, and Song show that import competition from China generated large and persistent worker-level adjustment costs. Workers initially employed in more exposed industries experienced lower cumulative earnings, weaker employment trajectories, more job churn, and greater reliance on SSDI. These costs were not evenly distributed: low-wage and low-tenure workers bore much larger burdens than high-wage workers.

8.2 Economic intuition

The mechanism is labor-market frictions. A China-induced demand shock reduces employment and earnings opportunities in exposed industries. If workers could instantly move to equally good jobs, initial industry exposure would not predict long-run earnings. But workers have industry-specific skills, firm-specific rents, search frictions, and possibly geographic constraints. As a result, the original shock follows workers through their subsequent employment histories.

8.3 Takeaway

The paper's central lesson is that trade adjustment is a worker-level distributional problem. Aggregate gains from trade can coexist with large, persistent, and unequal costs borne by workers initially tied to import-exposed industries. The most affected workers are often those with the least capacity to reallocate smoothly: low-wage workers, low-tenure workers, and workers with weaker pre-shock labor-force attachment.